

CSCE 670 INFORMATION RETRIEVAL · MODULE 05 · LECTURE NOTES

Evaluation

The course's centre: measuring a proxy for a judgement.

MODULE OUTCOMES

- Explain the test collection methodology and its assumptions.
- Build relevance judgements and assess their reliability.
- Choose a metric from how results will be used.
- Explain what click data gives and what it biases.
- Run a comparison whose result means something.

1 The methodology

A TEST COLLECTION IS A COLLECTION, A QUERY SET, AND RELEVANCE JUDGEMENTS — EACH WITH ASSUMPTIONS

The collection must resemble the documents your system will actually face — and **a benchmark collection from a different domain measures transfer rather than quality**, which is a different and usually harder question than the one you asked.

The queries must come from real information needs (Module 01 §4) — and **queries written by reading documents make the evaluation circular**, because the document is guaranteed findable by construction.

The judgements assume relevance is per-document, static, and independent of the session — **all three of which Module 01 §3 establishes are false** — and **the methodology accepts them as a working approximation**, which is defensible and should be stated rather than forgotten.

And completeness is unattainable: you cannot judge every document in a large collection, so unjudged documents are treated as non-relevant — **which systematically penalises a system that finds something the judging process missed**, and is the pooling bias below.

POOLING

run many different systems on each query, take the top k from each, and judge the union. Anything outside the pool is assumed non-relevant.

WHY IT IS NECESSARY

judging a million documents per query is simply impossible. The pool is a few hundred.

AND THE BIAS IT CREATES

a NEW system that retrieves a relevant document that no pooled system found is scored as if that document were irrelevant. So the collection systematically favours systems that resemble the ones which built the pool.

WHICH MATTERED when dense retrieval arrived: it found documents that lexical pools had missed, and the old collections consequently understated it.

SO: check the pool coverage for your own system before trusting an old collection's verdict on it.

Old test collections favour systems like the ones that built them — which is a structural bias rather than a flaw in any particular collection, and it is the methodology's deepest limitation. The practical check is cheap: take your system's top results that are unjudged, judge a sample of them yourself, and estimate how much the collection is understating you. **This is exactly the check that was missing during the early neural retrieval comparisons.**

2 Judgements

- **Write the criteria before judging anything** — because **criteria articulated afterwards describe what you happened to find** (Module 01 §4, and CSCE 676 Module 11's forking paths in a judgement setting).
- **Use graded rather than binary labels. Partially useful is the common case** (Module 01 §3), and a binary label discards exactly the information that distinguishes a good ranking from an adequate one.
- **Judge blind to which system retrieved the document**, and in a randomised order — both of which are easy to arrange and both of which are routinely neglected in small-scale evaluations.
- **Measure your agreement with yourself** by re-judging a sample a week later — **which is usually substantially lower than people expect**, and **bounds what any metric can resolve**: a difference smaller than your own inconsistency is not detectable.
- **And measure inter-judge agreement if you have two judges**, because a **metric difference smaller than the disagreement is not a finding**. Judge disagreement bounds the resolution of every comparison — and the reassuring empirical finding is that system *rankings* are fairly stable even when individual judgements are not, which is what makes the methodology workable at all.

3 Metrics

Metric	The user model it assumes	Use it when
Precision@k	The user looks at exactly k results and no further.	There is a fixed-size result page.
Recall	The user needs everything relevant and will keep looking.	Legal discovery, systematic review, patent search — where a miss is the expensive error.
Mean reciprocal rank	The user wants one answer and stops as soon as it is found.	Navigational queries (Module 01 §3).
Mean average precision	The user reads down the whole list, caring about every relevant document's position.	Binary judgements and a full ranking.
NDCG	Gain is graded, and discounted logarithmically by rank.	The general default — and the one to report unless you have a reason not to.

Every metric encodes a model of how the user reads the list — how far down they go, whether they stop, how much a lower position is worth — so choosing a metric is choosing a user model, and the choice should follow from Module 01 §3's intent taxonomy rather than from convention. The metric-as-user-model framing is this module's key idea, and it makes the choice arguable rather than arbitrary.

AND NO STANDARD METRIC ACCOUNTS FOR REDUNDANCY

Every metric in the table sums per-document gains — so ten identical relevant documents score exactly as well as ten different relevant ones, which is plainly wrong from the point of view of a person reading the list.

Which is Module 01 §3's point arriving as a concrete measurement failure: a list's relevance is not the sum of its documents' relevances, and the standard metrics assume that it is because the assumption makes them computable.

Diversity-aware metrics do exist — α -NDCG and the various subtopic measures — and they require subtopic judgements, which are considerably more expensive to produce than relevance judgements and are therefore rare.

So the practical answer is to measure diversity separately, with a simple pairwise-similarity statistic over the result list, and report it alongside the relevance metric — which is cheap and makes the limitation visible rather than hidden, which is this course's general move.

4 Clicks, and comparing

CLICKS ARE ABUNDANT AND POSITION-BIASED, SO THEY ARE RELATIVE EVIDENCE RATHER THAN ABSOLUTE

Users click higher-ranked results more often regardless of their relevance — the effect is large and well measured — so a **click-through rate is a measure of position as much as of quality**, and **comparing absolute click rates across differently-ordered systems is invalid**.

But a **click on a lower result when a higher one was skipped is informative** — the user saw both and chose the lower — **which is what the pairwise preference models extract**, and it is position-corrected by construction rather than by modelling.

And interleaving is the strong method: **mix two systems' results into a single list and observe which system's contributions receive the clicks** — **which controls for position directly and is far more sensitive than a conventional A/B test**, detecting differences with far less traffic.

With the important caveat that **clicks measure attractiveness: a misleading title gets clicked** — so **pair the click signal with a dwell time or an explicit satisfaction signal**, or you will optimise for bait (Module 12 §3's feedback argument).

- **Tune the baseline with equal effort** — a default-parameter BM25 against a carefully tuned system is **not a comparison** (Module 03 §3's callout, and CSCE 676 Module 13 §2).
- **Use enough queries. Fifty is a bare minimum and the variance across queries is large** — so **report a paired per-query significance test** rather than comparing two means.
- **Report the per-query distribution, not only the mean** — a system that improves the average while **hurting a third of the queries is a materially different proposition** from one that improves most queries slightly, and the mean cannot distinguish them.
- **Report the latency alongside the quality** (Module 04 §4), **because the two are traded** and a quality claim without a latency figure is incomplete.
- **And read the worst queries**, which is where the diagnosis is (Project 2's last requirement, and CSCE 638 Module 10 §4's error analysis). **The per-query distribution is the report that changes decisions** — a mean improvement concealing widespread regressions is a common and important finding, and it only shows up if you look.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapter 8 (free)	§1 and §3 — the metrics defined carefully, with the test collection methodology.
Sanderson — Test Collection Based Evaluation (free)	§1 and §2 critically — the assumptions, the pooling bias, and the judge agreement literature.

Resource	What it covers for this module
Chapelle et al. — Large-scale validation of click models (free)	\$4's position bias, measured — and the models that correct for it.
Radlinski & Craswell — Comparing the sensitivity of information retrieval metrics (free)	\$4's interleaving argument — how much more sensitive it is, with the numbers.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three assumptions** the judgements make, and which Module 01 says are false.
- 2 **Judge your system's unjudged top results** on an existing collection, and estimate the pooling bias against you.
- 3 **Write your relevance criteria**, then judge fifty query-document pairs.
- 4 **Re-judge ten of them a week later** and compute your agreement with yourself.
- 5 **Compute all five metrics** on the same run and compare the system rankings they imply.
- 6 **Construct a result list that scores well and is highly redundant**.
- 7 **Measure diversity separately** and report it alongside NDCG.
- 8 **Plot click rate against rank position** on any click log you can get.
- 9 **Run a paired significance test** between two systems over fifty queries.
- 10 **Plot the per-query difference** and report how many queries got worse.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three components of a test collection and an assumption each makes.
- 2 What is pooling, why is it necessary, and what bias does it create?
- 3 Why did the pooling bias matter for dense retrieval?
- 4 Give five rules for producing defensible judgements.
- 5 Why does judge disagreement bound every comparison?
- 6 Name five metrics and the user model each assumes.
- 7 What do all the standard metrics fail to account for?
- 8 Why are clicks relative rather than absolute evidence?

- 9 What does interleaving control for, and what do clicks still measure?
- 10 Give five requirements for a meaningful comparison.